# COVID-19 ICU Admission Prediction

Machine Learning Mini-Project Report | Logistic Regression using Baseline Clinical, Sex and CT Scanner Features

| | |
|---|---|
| **Objective** | Predict whether a COVID-19 patient is likely to require ICU admission. |
| **Dataset** | 81 patient records, 7 original columns; target: Intensive care unit. |
| **Model** | Logistic Regression with one-hot encoding of categorical features. |
| **Test split** | 80% training (64) and 20% testing (17), stratified, random_state = 42. |
| **Key result** | **Accuracy = 88.24%**; **ROC-AUC = 0.993**. |

## 1. Problem Statement

COVID-19 patient triage can benefit from early prediction of ICU admission so that high-risk patients can be identified from information available at baseline. The mini-project implements a supervised binary classification model that predicts **Non-ICU (0)** or **ICU (1)** using baseline patient information. The supplied reference study motivates this type of task by combining CT-derived/radiomics information with clinical and laboratory data for patient-outcome prediction.

## 2. Dataset Details

The notebook loads the supplied dataset description spreadsheet and reports **81 rows and 7 columns**: Patient ID, Sex, Baseline scanner, Follow-up Scanner, Baseline clinical score, Follow-up clinical score, and Intensive care unit. The target is converted from NO/YES to 0/1. The class distribution is **48 Non-ICU** and **33 ICU**. For the implemented model, the selected predictors are Baseline clinical score, Sex, and Baseline scanner.

| Original field | Role / observation |
|---|---|
| Baseline clinical score | Numeric predictor used by the model |
| Sex | Categorical predictor; one-hot encoded |
| Baseline scanner | Categorical predictor; one-hot encoded |
| Intensive care unit | Target: NO → 0, YES → 1 |
| Follow-up fields | Explored in the notebook, not used as model inputs |

## 3. Methodology and Implementation

- **Data preparation:** ICU labels are mapped from NO/YES to 0/1. Categorical inputs are converted using one-hot encoding with drop_first=True.
- **Feature matrix:** Baseline clinical score, Sex_M and Baseline scanner_Toshiba are the resulting model features.
- **Train/test split:** 80:20 stratified split gives 64 training samples and 17 testing samples.
- **Model:** Logistic Regression is fitted on the training set and used to generate class predictions and ICU probabilities.
- **Evaluation:** Accuracy, classification report, confusion matrix, ROC curve and ROC-AUC are calculated.

## 4. Results

| Metric | Result | Interpretation |
|---|---|---|
| Accuracy | 0.8824 (88.24%) | 15 of 17 test cases classified correctly |
| ROC-AUC | 0.9929 (~0.99) | Excellent ranking/separation on this test split |
| ICU precision | 1.00 | All predicted ICU cases were actually ICU |
| ICU recall | 0.71 | 5 of 7 actual ICU cases were detected |
| ICU F1-score | 0.83 | Balance of ICU precision and recall |

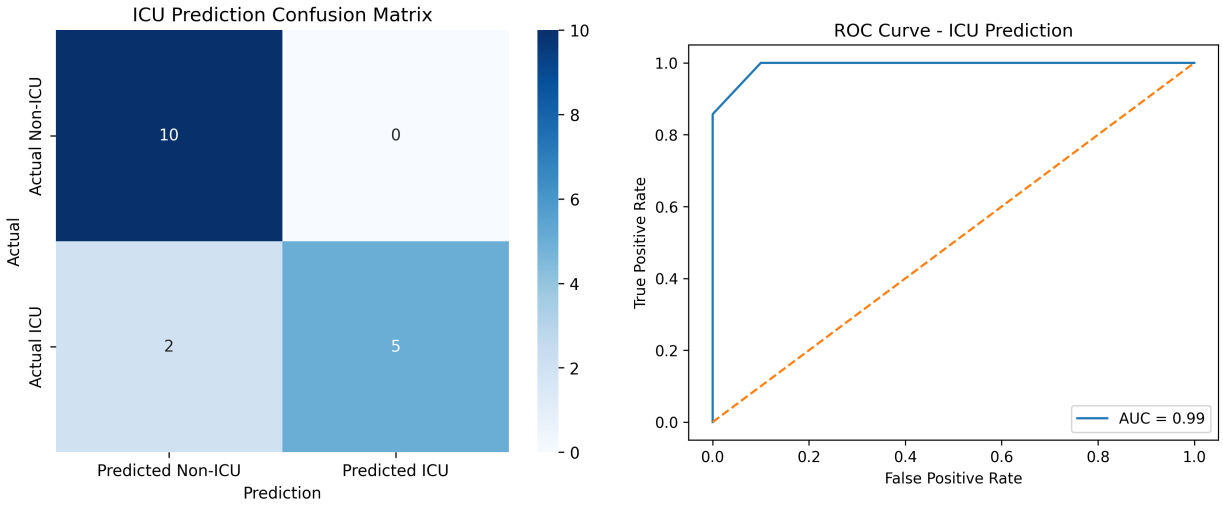



Figure 1. Confusion matrix and ROC curve generated by the notebook. The confusion matrix is [[10, 0], [2, 5]] and the ROC-AUC is approximately 0.993.

# COVID-19 ICU Admission Prediction — Detailed Summary

Implementation, interpretation, conclusion and demonstration plan

## 5. Interpretation of the Confusion Matrix

The test set contains 17 observations. The confusion matrix has **10 true negatives**, **0 false positives**, **2 false negatives**, and **5 true positives**. Thus, the model did not incorrectly label any Non-ICU patient as ICU in this test set, but it missed two actual ICU cases. This is important for a triage application because recall for the ICU class is especially relevant.

## 6. Feature / Model Information

| Feature | Coefficient | Meaning in the fitted model |
|---|---|---|
| Baseline clinical score | +1.2306 | Higher score increases the model's ICU probability |
| Sex_M | +0.3971 | Male category has a positive fitted coefficient relative to the dropped category |
| Baseline scanner_Toshiba | –1.0182 | Toshiba category has a negative fitted coefficient relative to the dropped category |

## 7. Conclusion

The mini-project successfully demonstrates an end-to-end machine-learning workflow for binary ICU admission prediction: data inspection, target encoding, categorical feature encoding, stratified train/test splitting, Logistic Regression training, prediction, and performance visualization. On the available 17-case test split, the model achieved **88.24% accuracy** and **0.9929 ROC-AUC**. The results show strong performance on this small dataset, while the two missed ICU cases show that further work is needed before any real clinical use.

## 8. Limitations and Future Scope

- The notebook uses only 81 records, so the test set is small (17 cases); performance estimates may vary substantially with a different split.
- The implemented model uses baseline clinical score, sex and baseline scanner rather than extracted radiomics features.
- The provided reference study uses a much richer multi-source setting (radiomics + clinical + laboratory information) and much larger cohorts.
- Future work can evaluate additional clinically relevant features, compare multiple classifiers, use cross-validation, address class imbalance where necessary, and validate on an independent cohort.

## 9. Live Demonstration Plan

| Step | What to show |
|---|---|
| 1 | Load dataset and display shape / columns |
| 2 | Map ICU target to 0/1 and show class distribution |
| 3 | Build X and y with the selected three predictors |
| 4 | Show stratified 80:20 split and train Logistic Regression |
| 5 | Show accuracy, classification report and confusion matrix |
| 6 | Show ROC curve, AUC and final results table |

## References / Source Materials

[1] Mini-Project Assignment Guidelines, UE24CS352A – Machine Learning, 2026. The assignment requires a concise PDF write-up covering problem statement, dataset, approach, implementation and conclusions, plus a presentation and live demonstration.

[2] Zhan, Xianghao; Li, Yiheng. “CT-based Patient Triage of COVID-19: Radiomics Prediction of ICU Admission, Mechanical Ventilation, and Death of Patients.” Provided reference PDF. The study reports radiomics, clinical and laboratory data fusion and compares multiple machine-learning models.

**Scope note:** The notebook also contains an initial artificial-data practice block. The reported project results in this document come from the later 81-patient dataset workflow in the notebook, not from that artificial-data block.